DATA ANALYTICS LEVEL III

Day 6: Introduction to Databases & SQL — Your First Queries

Student Activities

How to Use This Document

This is your activity workbook for today's 4-hour meeting. Activity 1 is guided — we set up the tools together. Activity 2 is your first solo querying session. For every query question, write (or paste) BOTH the SQL you ran AND the answer it returned.

Activity	What You Submit	Time
Activity 1: Set Up & Connect	Screenshot of your first query result	40 min
Activity 2: Your First Queries	Your queries + answers (this doc or a .sql file)	60 min
Learning Journal Entry #7	Your journal reflection	20 min

Activity 1 — Set Up & Connect (Guided)

Goal: Install the SQL toolkit, connect to the practice database, and run your very first query.

Step 1 — Install the extensions

1. Open VS Code → Extensions (Ctrl+Shift+X)
2. Search “SQLTools” (by Matheus Teixeira) → Install
3. Search “SQLTools SQLite” → Install

Step 2 — Get the database

4. Download Day06_practice.db from the LMS
5. Save it in a folder you can find again (e.g., Documents\SQL)

Step 3 — Connect

6. Ctrl+Shift+P → “SQLTools: Add New Connection” → SQLite

7. Connection name: Day06 | Database file: browse to Day06_practice.db
8. TEST CONNECTION → SAVE CONNECTION → CONNECT NOW

Step 4 — Your first query

In the SQL file that opens, type the query below, highlight it, and press Ctrl+E Ctrl+E:

```
SELECT *
FROM employees;
```

- You should see 20 rows. Take a screenshot of the result panel.
- **Now try:** SELECT * FROM products; — how many rows does it return?

Submit: Day06_Activity1_[YourName].png (screenshot of your first query result)

Activity 2 — Your First Queries

Goal: Answer real questions from the database using SELECT and WHERE.

For each question: write the SQL query, run it, and record the answer (row count, names, or values as asked).

Part A — SELECT (choosing columns)

9. Show ALL columns and rows of the employees table. How many rows are there?
10. Show only first_name, last_name, and department for all employees.
11. Show only product_name and price for all products.

Part B — WHERE (filtering rows)

12. Which employees work in Manila? (Show first_name, last_name, city.) How many are there?
13. Which employees earn MORE than ₱40,000? (Show first_name and salary.) How many?
14. List everyone in the IT department. (Show first_name and position.)
15. Which products cost ₱1,000 or LESS? (Show product_name and price.)
16. Which products have FEWER than 10 units in stock? (Show product_name and stock_qty.) These need re-ordering!

Part C — Challenge

17. Using <> (not equal), count how many employees do NOT work in Makati. (Hint: SELECT COUNT(*) …)
18. Managers earn ₱50,000 or more in this company. Write ONE query that shows first_name, last_name, and salary of everyone earning at least ₱50,000.

Submit as: Day06_Activity2_[YourName].docx (or a .sql file with comments)

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Learning Journal Entry #7

Goal: Reflect on your first day of SQL.

1. In your own words: what is a database, and how is it different from a spreadsheet?

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2. Explain what each part of this query does: SELECT first_name FROM employees WHERE city = 'Cebu';

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3. What surprised you most about SQL today?

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4. My confidence with SQL today (1 = lost, 5 = very confident), and why:

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Submit as: Day06_Journal7_[YourName].docx

Submission Checklist

- **[ ] Activity 1:** screenshot of your first query result (connected + 20 rows)
- **[ ] Activity 2:** all 10 queries with their answers
- [ ] Learning Journal Entry #7